

THE MILLENNIUM PROBLEMS

BY

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RECONNAISSANCE

THE CORPUS, THE STATE OF THE FIELD, AND WHAT THIS TOOLKIT REACHES

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*For Megan, my loving wife, the only person that really understands
what I mean when I speak.*

*For Autumn and Opal, without you this thought would have never
occurred.*

To everyone else, I wish you well.

ABSTRACT

This is a reconnaissance and states no result. It records what the Millennium Prize problems ask, in the words of their official statements; what was published against one of them in September 2026; which of the rest an arbitrary-precision apparatus could be aimed at; and the reading that is not done. No problem is attacked here, and no priority is claimed over anything.

The corpus is seven documents. Five are Clay Mathematics Institute problem descriptions, covering P versus NP, the Hodge conjecture, the Riemann hypothesis, quantum Yang–Mills theory, and the Birch and Swinnerton-Dyer conjecture. Two are Riemann: his 1859 memoir in the Wilkins translation, and four photographed pages of his handwriting. Each was identified from its own pages. The photographs hold a working draft and not the published memoir, carry no extractable text, and circulate under a filename that names the memoir; a citation built from that filename would point at the wrong document.

Navier–Stokes is absent from the corpus because the Clay index of unsolved problems does not list it. On 8 September 2026 OpenAI published a proof of finite-time blowup for the forced three-dimensional incompressible equations, with a Lean 4 formalization. Fefferman’s statement asks for a proof of any one of four statements, and two of the four name a smooth external force. Forcing is therefore inside the problem as posed.

The formalized statement is checked against the problem statement on six points: viscosity strictly positive with no vanishing limit, the decay conditions on the data and on the force, a force smooth on the closed half-line, a conclusion quantified as a negated existential, kinetic energy bounded for all time, and the periodic pressure condition the errata add. All six match. The statement is not the authors’ own: it is adapted from the Formal Conjectures project’s published formalization, and a separate module discharges it. A proof assistant checks a deduction and never a translation, and this arrangement answers that objection as far as an arrangement can.

The limits are stated where the claims are. The Lean development is not built and nothing here is machine-checked. Of the papers, the statements and the bridging argument are read and the proofs are not. A priority dispute runs in this field and no position is taken on it.

The problem the apparatus can reach is Birch and Swinnerton-Dyer. The refined conjecture is an exact numerical identity, and checking an instance comes down to whether a computed real number lands on an integer. The Riemann hypothesis is the trap: the apparatus matches its subject and not its obstruction, and counting more zeros bears on no proof. The null for the Birch and Swinnerton-Dyer measurement is drawn through the same pipeline as the effect, because a derived null sets the bar too low.

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WHAT THIS IS

This chapter states what this research paper records and the boundary of what it claims.

1.1 THE RESEARCH PAPER

The aim is narrow: check one specific claim against its source documents and record what the check finds.

Four things are in it. First the corpus, and what each document is once opened, as against what its filename promises. Then the state of the Navier–Stokes problem as of September 2026. Then a reading of the published blowup result against the official problem statement and against the Lean formalization published with it. Last, an assessment of where an arbitrary-precision apparatus could be aimed among the remaining problems.

No problem is attacked. Nothing is measured, run, or begun. Leading a research paper with a method claims the method reaches something; no claim of that kind appears here or is supported here.

1.2 WHAT IS NOT CLAIMED

No priority is claimed. The crystallography research paper [17] says the same of its own construction.

The reading of the literature is not done. For any of the five open problems in the corpus, somebody has to work through the field behind it and set down what that field already knows. Until then, these pages assert no new method and credit no old one.

A priority dispute is running in the field the Navier–Stokes chapter covers. It is not investigated here, no position is taken, and no line in that chapter is evidence on either side. The chapter reports what the documents say and does not ask who reached the result first.

1.3 METHOD

Five rules govern the text, and each limits what the later chapters may claim.

Cite nothing unread. A document is cited here only once it is opened. A fact that arrives by report is marked as reported in the sentence that carries it.

A secondhand fact is no second witness. A fact that arrives secondhand is confirmed against the document before it is written down.

Gaps go in the sentence making the claim. Not a footnote, not an appendix. The Navier–Stokes chapter has a section on what is not verified, and marks the individual soft spots where they fall.

A claim that does not hold stays on the page, with what refutes it. The derived null in the toolkit chapter is kept beside the drawn null that replaces it.

Draw the bar. Never derive it. A threshold reasoned out of an assumed distribution omits every variance source not enumerated, and omitted variance only ever pushes one way. The toolkit chapter applies the rule to the recommended measurement.

THE CORPUS

This chapter states what each document in the corpus is, as established by opening it, and separates that from what its filename says. The corpus is seven files, 7,852,267 bytes.

2.1 WHAT IS IN IT

Each document is identified from its own first page and embedded metadata, not from its filename. Five are the official Clay Mathematics Institute problem descriptions [4, 6, 2, 11, 21]. Two are Riemann [22, 19].

File	Identified as	Pages	Bytes
pvsnp.pdf	Stephen Cook, <i>The P versus NP Problem</i>	12	176,204
hodge.pdf	Pierre Deligne, <i>The Hodge Conjecture</i>	5	146,894
riemann.pdf	E. Bombieri, <i>Problems of the Millennium: the Riemann Hypothesis</i>	11	159,267
yangmills.pdf	Jaffe and Witten, <i>Quantum Yang–Mills Theory</i>	14	195,976
birchswin.pdf	Andrew Wiles, <i>The Birch and Swinnerton-Dyer Conjecture</i>	5	142,908
Wilkins-translation.pdf	Riemann 1859, <i>On the Number of Prime Numbers less than a Given Quantity</i> , translated by David R. Wilkins	10	119,504
riemann1859.pdf	Not the 1859 paper. See below.	4	6,911,514

2.2 THE CORPUS IS COMPLETE

Six Millennium problems are commonly counted as open, and the corpus holds five official statements. The natural first reading is that Navier–Stokes is missing. That reading is wrong.

The Clay Mathematics Institute index of Millennium Prize problems [3], accessed 11 September 2026, lists five problems as unsolved, Birch and Swinnerton-Dyer, Hodge, P versus NP, Riemann, and Yang–Mills and the mass gap, and one as solved, Poincaré. Navier–Stokes appears in neither list. The corpus holds exactly the five the index lists as unsolved.

That is a fact about how a website categorizes its own contents on one date at one address, and it is recorded as that and not as a fact about the status of a problem. A site can be mid-edit, deliberately cautious, or behind. The Navier–Stokes problem’s own page, accessed the same day, carries the status label *Active*. The two readings are not consistent with each other, both are recorded, and neither is treated here as evidence about the mathematics.

2.3 THE DOCUMENT THAT IS NOT WHAT IT IS NAMED

`riemann1859.pdf` is not Riemann’s 1859 memoir. It is a photographic scan of Riemann’s handwritten working manuscript [19].

The evidence is in the file. Its four pages carry zero extractable characters between them, measured page by page. The embedded document title is a scanner path, `Cod_Ms_Riemann_3.jpg`, under a directory named for the Riemann *Nachlass* and a subdirectory named *Rechenseiten*, calculation pages. The producer chain is a JPEG passed through PDF24 Creator and a Quartz PDF context in 2015, which is a scan-to-PDF path and not a typesetting one. Rendered and read as images, all four pages are German cursive in ink, heavily struck through with interlinear corrections, foxed and water-stained, with *Nachlass Riemann* and the year 1859 legible in the header of the first page. The zeta function, the factor $\Pi(s-1)$, the logarithmic expansion of the Euler product, the function $\xi(t)$ and the logarithmic integral $\text{Li}(x)$ are identifiable in the hand.

The file is a primary source in the most literal available sense, and it is unusable for text search or for quotation without a reader of nineteenth-century German *Kurrentschrift*. The content of the published memoir is in the Wilkins translation [22] instead.

A citation built from this filename is wrong. It would name Riemann’s published memoir and point at a manuscript draft. A bibliographic entry left with empty fields until somebody looks the source up does not catch the error, because the error is in the identification and not in the completeness. The distinction has to be written into the entry, and it is in this paper’s.

2.4 AN OBSERVATION ABOUT PRODUCTION

Five of the six official statements, counting the Navier–Stokes statement [8] fetched separately for the next chapter, share a producer string of MiKTeX pdfTeX-1.21a and creation timestamps on 2026-08-04 within three minutes of one another; the Navier–Stokes file is stamped 16:46:57, nineteen seconds after the Birch and Swinnerton-Dyer file. Bombieri’s Riemann statement comes from older tooling, a dvips and Acrobat Distiller chain dated 2000-08-30, and carries the embedded title `Riemann-new.dvi`.

The observation is about how the statements are produced and says nothing about their contents. It establishes that the corpus and the separately fetched statement are the same edition of the same set, which matters for a bibliographic entry and for no argument about mathematics.

NAVIER–STOKES, AND WHAT A FORMALIZATION CHECKS

This chapter states what the official problem asks, what was published against it on 8 September 2026, and the result of checking the published formalization’s statement against the problem statement point by point. The documents are Fefferman’s Clay problem description [8], 6 pages; the Navier–Stokes paper [15], 166 pages; the Euler paper [13], 57 pages; and the Lean 4 repository [14], of which four source files are read, one of them the adapted Formal Conjectures statement [9].

3.1 WHAT THE PROBLEM ASKS

Fefferman’s official description sets up the incompressible equations on $\mathbb{R}^n$ for $n = 2$ or 3 , with viscosity $\nu > 0$, and imposes decay and regularity conditions numbered (4) through (11). It then states, in the sentence that governs everything after it:

To give reasonable leeway to solvers while retaining the heart of the problem, we ask for a proof of one of the following four statements.

The four are (A) existence and smoothness on $\mathbb{R}^3$, (B) the same on $\mathbb{R}^3/\mathbb{Z}^3$, (C) breakdown on $\mathbb{R}^3$, and (D) breakdown on $\mathbb{R}^3/\mathbb{Z}^3$. In (A) and (B) the force f is taken identically zero. In (C) and (D) it is not. Statement (C) reads, in his words, that for $\nu > 0$ and $n = 3$ there exist a smooth divergence-free u° on $\mathbb{R}^3$ and a smooth $f(x, t)$ on $\mathbb{R}^3 \times [0, \infty)$, satisfying (4) and (5), for which there exist no solutions (p, u) of (1), (2), (3), (6), (7) on $\mathbb{R}^3 \times [0, \infty)$. Statement (D) is the same with (8) and (9) on the data and (10) and (11) on the solution.

A smooth external force is therefore named inside two of the four statements, and the problem asks for a proof of any one of the four. The only occurrence of the word *prize* in the document is a remark that the Euler equations are not on the Clay Institute’s list. There is no passage reserving the problem for (A) or (B).

The conditions that matter for the check below are (4), rapid decay of every derivative of the initial data; (5), rapid decay of every space-time derivative of the force on the closed half-line; (6), smoothness of p and u on $\mathbb{R}^n \times [0, \infty)$; and (7), $\int |u(x, t)|^2 dx$ bounded by one constant for all $t \geq 0$, which Fefferman annotates *bounded energy*.

The document carries errata and one of them adds a condition. The first reads that the further condition $p(x + e_j, t) = p(x, t)$ should be made explicit in Eqn (8). The periodic problem therefore requires a periodic pressure, which the body of the statement does not say.

3.2 WHAT WAS PUBLISHED AGAINST IT

On 8 September 2026 OpenAI published *Finite time blowup for Navier–Stokes* [15], 166 pages, together with a companion Euler paper and a Lean 4 repository. Its Theorem 1.1 is a construction, not directly a non-existence statement:

For every $\nu > 0$ there exist a force $f \in C_c^\infty(\mathbb{R}^3 \times (0, 1); \mathbb{R}^3)$, a compact set $K \subset \mathbb{R}^3$, and smooth velocity and pressure fields u, p on $\mathbb{R}^3 \times [0, 1)$ satisfying $\partial_t u + (u \cdot \nabla)u - \nu \Delta u + \nabla p = f$, $\nabla \cdot u = 0$, $u(\cdot, 0) = 0$, such that $\text{supp } u(\cdot, t) \cup \text{supp } p(\cdot, t) \subseteq K$ for every $0 \leq t < 1$, $\sup_t \|u(t)\|_{L^2(\mathbb{R}^3)} < \infty$, and $\limsup_{t \rightarrow 1} \|u(t)\|_{L^\infty(\mathbb{R}^3)} = \infty$.

Constructing one solution that blows up does not by itself establish (C). Statement (C) asks that *no* smooth bounded-energy solution exist on the whole half-line for that datum and force, and a blowing-up solution leaves open whether some other smooth global one exists. The paper closes that gap and says so in the sentence after the theorem, asserting that there is then no smooth solution on $\mathbb{R}^3 \times [0, \infty)$ with the same force and initial datum whose kinetic energy is uniformly bounded, and that this establishes alternative (C) as stated by Fefferman.

The argument for the gap is in the proof of Theorem 1.1 in section 10.4. Smoothness and fixed compact support put the field in $C([0, T]; H^3)$ for each $T < 1$; the embedding $H^3(\mathbb{R}^3) \hookrightarrow L^\infty(\mathbb{R}^3)$ excludes a classical H^3 continuation through time one; a difference-energy calculation with Grönwall gives uniqueness on every shorter interval, making $[0, 1)$ the maximal classical existence interval. A supposed global smooth competitor of uniformly bounded energy is then forced by the comparison lemma to agree with the constructed field throughout $[0, 1)$, which contradicts its own boundedness. Every fixed positive viscosity is reached by a spatial rescaling that leaves the singular time at one.

Statement (D) is Corollary 10.6, obtained by rescaling the compact supports into the interior of a fundamental cube and summing integer translates, whose supports stay disjoint so the nonlinear term survives. The corollary’s proof states that the pressure is periodic as well, *as required in the erratum to the problem statement*. The paper accounts for the errata.

3.3 WHAT A PROOF ASSISTANT DOES NOT CHECK

The position taken here on machine-checked proof is that Lean verifies the deduction and never the claim. A proof assistant confirms that a conclusion follows from the assumptions as written. It cannot confirm that the statement formalized is the statement anybody meant. A machine-checked proof of a neighboring theorem compiles exactly as cleanly as a machine-checked proof of the right one, and nothing in the compile distinguishes them. The burden therefore moves entirely onto the formalized statement, and that statement is a translation. No proof assistant checks a translation.

That makes the formalized statement a finite document a person reads word by word. The rest of this chapter is that reading.

3.4 THE STATEMENT CHECK

Six points are checked against the Lean statement in `ComparatorChallenges/NavierStokes.lean` and against Fefferman’s numbered conditions. The first five are specified before the file is opened. The sixth is added because the file itself raises it.

Point	What the Lean statement carries	Verdict
Viscosity strictly positive, and not a vanishing-viscosity limit	$(\nu : \mathbb{R}) (\text{h}\nu : \nu > 0),$ universally quantified over the theorem	matches
Condition (4) on the initial data, unrelaxed	<code>InitialVelocityConditionDecay</code> : divergence-free, C^∞ , and for every derivative order m and every K a constant C with $\ \nabla^m u_0(x)\ \leq C/(1 + \ x\)^K$	matches
Condition (5) on the force, unrelaxed, on the closed half-line	<code>ForceConditionDecay</code> : C^∞ on $\text{univ} \times [0, \infty)$ and $\ \nabla^m f(x, t)\ \leq C/(1 + \ x\ + t)^K$ for all $t \geq 0$	matches
Conclusion in the negated existential form Fefferman quantifies	<code>not (exists v p, NavierStokesExistenceAndSmoothnessRn nu u0 f v p)</code>	matches
Energy bounded uniformly in time, not at $t = 0$ alone	<code>globally_bounded_energy</code> : there exists E such that for all $t \geq 0$, $\int \ v(x, t)\ ^2 < E$; plus square-integrability at each $t \geq 0$	matches
The periodic pressure condition from the errata	<code>isOnePeriodic_pressure</code> on the periodic structure, annotated in the source as following the Clay errata	matches

The equation itself is encoded with the time derivative taken as a derivative within $[0, \infty)$. Fefferman states equation (1) on the closed half-line, and the Lean encoding follows him there. Divergence is defined as the trace of the Jacobian and carries an explicit junk value of zero at points of non-differentiability, which the source declares instead of leaving implicit.

3.5 THE STATEMENT IS NOT THE AUTHORS’ OWN

The file carrying the statement is a *comparator challenge*. Its header records that it is copied, at a named commit, from the Formal Conjectures project’s Lean formalization of the Navier–Stokes problem statement [9], that the breakdown alternatives (C) and (D) are retained *including their intentional placeholder gaps*, and that neither the proof root nor the submission imports it. Both theorems in that file end in a placeholder and prove nothing. It is a yardstick.

The discharge is a separate module. It restates both theorems with textually identical statements and closes each by applying a bridge lemma from the project’s own development, declaring no axioms and leaving no placeholder, and it ends by printing the axiom dependencies of both theorems. The repository’s instructions direct a checker to verify the two challenges with the Comparator

tool under a sandbox, using an independent exporter and an independent kernel instead of Lean’s own [14].

That architecture answers the objection this chapter opens with. The concern is that an author who writes both the statement and the proof can machine-check a translation of their own choosing. Here the statement is written by a third party, published beforehand, for the purpose of stating the problem and not of being proved; the authors adopt it unmodified and discharge it; and the axiom base is printed for a reader to check. The translation is still the thing to read, and this chapter reads it, but the party that wrote it has no stake in what is later proved against it.

3.6 THE UNFORCED EULER COMPANION

A second paper is published alongside, *Finite time blowup for the Euler equation* [13], 57 pages, and it matters here for one reason: it is unforced.

It studies the unforced incompressible Euler equations on $\mathbb{R}^3$, $\partial_t u + (u \cdot \nabla)u + \nabla p = 0$ with $\nabla \cdot u = 0$, taking initial data in the smooth compactly supported divergence-free class and writing $T(u_0)$ for the maximal lifespan of the smooth solution. Its Theorem 1.1 asserts that there exists such a u_0 with $0 < T(u_0) < \infty$, whose smooth solution satisfies $\limsup_{t \rightarrow T} \|u(t)\|_{L^\infty} = \infty$ and $\int_0^T \|\operatorname{curl} u(t)\|_{L^\infty} dt = \infty$.

The second divergence is not decoration. The Beale–Kato–Majda criterion, which the paper states and attributes, says a smooth three-dimensional Euler solution continues past T whenever that integral is finite. Finite-time breakdown therefore requires it to diverge, and the theorem delivers exactly the quantity the criterion demands. By the authors’ own account the Euler result comes first, as the easier problem, and is used to prompt the Navier–Stokes effort.

This is not a Millennium result and is not recorded as one. Fefferman’s statement says the problems are also open and very important for the Euler equations at $\nu = 0$, and that the Euler equation is not on the Clay Institute’s list of prize problems. The distinction is his and is kept here.

3.7 WHAT THE UNFORCED EULER RESULT DOES NOT SETTLE

The question the Euler paper raises is what is left as the forcing goes to zero. The test behind it is whether a mechanism repairs a property of the object or a property of the setup; a mechanism that needs the force to be there teaches something about forced systems and nothing about the unforced problem.

The Euler paper looks like the answer and is not. **Removing the force and removing the viscosity are different removals.** Statements (A) and (B), the two alternatives that remain open on anybody’s reading, are $\nu > 0$ with f identically zero. The Euler equations are $\nu = 0$. An unforced blowup at zero viscosity therefore does not transfer, and it does not transfer for the precise reason that the regularity problem exists: the viscous term is the smoothing whose sufficiency is the entire question. A construction that blows up with no viscosity to fight is not evidence about what happens when there is.

Put as the limit test, the two papers move in orthogonal directions. One removes the viscosity and keeps nothing to resist the amplification. The other keeps the viscosity and supplies a force. The open cases keep the viscosity and remove the force, and neither paper is that case.

That is an argument from the shapes of the statements and not from their mechanisms. The amplification mechanism is not read in either paper, and the full version of the limit test requires reading it instead of reasoning from the abstract. What is recorded here is that the question is still open and that the obvious shortcut to it does not work.

3.8 WHAT IS NOT VERIFIED, STATED HERE AND NOT IN A FOOTNOTE

Nothing here is machine-checked. The Lean development is not built. No proof is executed, no axiom output is observed, and the claim that the discharge module compiles without placeholders rests on reading its source text. The check in this chapter is a reading of four files out of 2,669.

The proofs are not read. The 166-page paper is read at its abstract, its introduction, its Theorem 1.1, the section 10.4 argument that bridges to the non-existence form, and Corollary 10.6. The mathematical content of sections 3 through 9 and the three appendices is not read. The bridge lemma the discharge module applies is not traced into the development. Of the Euler paper's 57 pages, only the abstract, the introduction and Theorem 1.1 are read; its amplification mechanism, the part the limit test needs, is not. Nothing in this chapter rests on the correctness of either proof, and nothing in it could, because neither is read.

The priority dispute is not investigated. A dispute over credit is running in this field. It is not examined here, no position is taken on it, and nothing in this chapter bears on it either way.

One fact rests on press reporting. That OpenAI does not intend to claim the Millennium Prize is reported in the press [20] and is not confirmed here against a statement of its own. It is recorded as reported.

And the status question is separate from the mathematics question. Whether the Clay Mathematics Institute accepts, awards, or will award anything is not verified here and is not asserted. What this chapter establishes is narrower and is the part that is checkable: the statement discharged in Lean is the statement Fefferman wrote, on all six points examined.

WHAT THIS TOOLKIT REACHES

This chapter assesses which problem an arbitrary-precision apparatus could be pointed at, the reasoning for it, the trap beside it, and the measured bound that constrains it. The assessment is a recommendation and no work on it is begun.

4.1 THE OBVIOUS ANSWER, AND WHY IT IS THE TRAP

An apparatus whose edge is arbitrary precision reaches for Riemann first, and Riemann is the wrong choice for a reason that has nothing to do with how good the arithmetic is.

The apparatus matches the *subject* and not the *obstruction*. A zeta function and unbounded precision look like a fit. But zeros on the critical line are computed far past anything a desktop reaches. The first 10^{13} zeros are located on the line [10], zeros near the 10^{22} nd are computed [12], and the hypothesis is verified rigorously up to height $3 \cdot 10^{12}$ [16]. No finite count of zeros bears on the proof at all. Adding zeros reproduces existing work at a worse scale. The conjecture is not precision-bound, and an apparatus whose advantage is precision buys nothing against it.

That is the superficial-match failure in its exact form, and it is worth naming because it is the shape of mistake this apparatus is most exposed to: the subject looks like the instrument. Yang–Mills wants constructive quantum field theory and lattice gauge machinery. Hodge wants algebraic geometry with no numerical handle this toolkit has. P versus NP is not a numerical question.

4.2 THE RECOMMENDATION

Birch and Swinnerton-Dyer, because its obstruction matches the instrument, where the other Millennium problems match only in subject.

It is the only one of them found by computation: Birch and Swinnerton-Dyer fitted curves numerically on EDSAC [1]. It is natively experimental, and the refined conjecture is an exact numerical identity [21]. For an elliptic curve E of rank r , writing Sha for the order of the Tate–Shafarevich group, Ω for the real period, Reg for the regulator and c_p for the Tamagawa numbers,

$$\frac{L^{(r)}(E, 1)}{r!} = \frac{\text{Sha} \cdot \Omega \cdot \text{Reg} \cdot \prod_p c_p}{|E_{\text{tors}}|^2}.$$

Rearranged, the analytic order of Sha is a quotient of computable real numbers, and the conjecture asserts that this quotient is an integer, usually a perfect square. The verification is therefore: compute a real number to N digits and ask whether it lands on an integer. That is a precision-bound question, and it is the check the arithmetic already performs.

4.3 THE FIRST MEASUREMENT, AND THE POSITIVE CONTROL IT REQUIRES

The positive control comes first and is not optional. Curves are taken from published tables [5] where the rank and the order of Sha are already known, at rank zero and one, small conductor, with Sha trivial or known to be 4 or 9. The quotient is computed and must land on the known integer. An instrument that cannot reproduce a known answer says nothing when pointed at an unknown one.

The quotient is only as good as its weakest input. The real period and the regulator come from their own algorithms carrying their own error. The report is therefore the precision of the quotient and not the precision of the arithmetic. An arithmetic with no floor under it does not give its inputs one.

4.4 THE NULL, DRAWN AND NOT DERIVED

The derived null does not hold. It assumes that a real number carrying no integrality structure has its distance to the nearest integer distributed uniformly on $[0, \frac{1}{2}]$, that landing within 10^{-N} of an integer by chance therefore has probability about 2×10^{-N} , and that agreement to fifty digits is a bound of 10^{-50} .

What refutes it. It is derived and not drawn. A threshold reasoned out from an assumed distribution omits the variance sources not enumerated, and omitted variance only ever pushes one way.

One narrower objection to it is itself wrong. It holds that the integer factors in the quotient, the torsion order squared, the Tamagawa product and $r!$, might concentrate the quotient near integers and inflate the coincidence rate. They do not: scaling a structureless real by a known rational does not concentrate it near integers. The derived null fails for the better reason and not for that one.

The drawn null measures the null under the same conditions as the effect, through the same pipeline and with the same inputs and the same precision losses, in two ways. Forming the quotient with the rank taken one too high or one too low gives a condition under which the identity should not hold, and the distribution of distance-to-nearest-integer across many curves is then the null itself. Computing the quotient across many curves whose Sha is known gives the spread of the residual about the true integer. That spread is the instrument's own floor, and the period and the regulator dominate it well before the arithmetic does. The second is the positive control and the floor measurement in one pass.

The consequence changes what may be claimed. The bound is not 10^{-N} for N digits of arithmetic. It is however close the pipeline gets on a curve whose answer is already known. If the regulator runs out at 10^{-12} , then 10^{-12} is the claim however many digits the multiply carries.

4.5 THE MEASURED BOUND THAT CONSTRAINS THIS

The cost of depth in the integer relation search of the same arithmetic [18] is measured, and the measurement constrains the plan.

Cost goes as precision to the power 2.64. The prediction is 2.585, from a Karatsuba host multiply near width^{1.585} against a count linear in the width, and the reduction step does not account for the difference.

The device overtakes the host at 1,024 limbs, 32,768 bits. The mean operand runs at 0.642 of the precision, and the mean multiply reaches that crossover near 51,000 binary places. One cell at 51,036 places costs 26.2 hours.

None of that binds the recommended measurement, which wants tens to hundreds of digits. It binds two other things: relation detection at tens of thousands of digits is out of reach, and so is any plan that relies on the device to rescue it.

4.6 WHAT IS NOT CLAIMED

This is verification and not proof, and that distinction governs everything claimed in this chapter.

A measurement of this kind produces evidence for particular instances of the conjecture, in the way the original EDSAC work did. A verified instance at unusual rank or unusual precision is a real contribution and it is not a solution.

The recommendation carries plain caveats. No expertise in elliptic-curve L -function computation stands behind it. Computing $L^{(r)}(E, 1)$ requires modular symbols [5] or Dokchitser's algorithm [7], established systems already implement those, and rebuilding them would be wasted effort. The realistic shape is established algorithms for the L -function with this arithmetic supplying precision underneath, and the first task is finding out whether that seam works at all.

SOURCES, AND HOW MUCH OF EACH WAS READ

This chapter names every source the research paper rests on and states how much of it is read, so that no claim here traces to a document nobody opened.

A surname in a sentence buys a reader nothing; an author, a title and an identifier they can follow buy them the source. Fields a document does not carry stay empty, with a note naming which. The extent of reading belongs beside each entry. Five sources are opened to the first page, two are read in part with the part named, one is read in full, and one cannot be read as text. Without that, a bibliography degrades into a list of things that exist.

5.1 READ IN FULL

Fefferman [8], *Existence and Smoothness of the Navier–Stokes Equation*. Six pages, errata included, read cover to cover; the Navier–Stokes chapter turns on it. No year appears anywhere on it, and the paper examined in that chapter cites it without one too; the entry carries none. Its bytes are identical to the copy claymath.org serves, accessed 11 September 2026.

5.2 READ IN PART, WITH THE PART NAMED

Finite time blowup for Navier–Stokes [15]. 166 pages. Abstract, introduction, Theorem 1.1, the bridging argument in section 10.4, Corollary 10.6. Sections 3 through 9 and all three appendices carry the mathematics and are not read.

Finite time blowup for the Euler equation [13]. 57 pages. Abstract, introduction, Theorem 1.1. The amplification mechanism is in the unread remainder, and the limit test in the Navier–Stokes chapter wants that mechanism.

Formal Conjectures [9], the Lean formalization of the Navier–Stokes problem statement. Read as source text, in the adapted copy the Lean repository [14] carries, at the commit that file names. Four files out of 2,669, none of them built. Everything this research paper finds about the formalized statement rests on that one file.

5.3 READ AT THE FIRST PAGE ONLY

Five official Clay problem descriptions, for the five problems the index lists as unsolved. Each is opened, identified from its own first page instead of its filename, and closed. All five are byte-identical to the copies claymath.org serves, accessed 11 September 2026.

Cook [4], *The P versus NP Problem*, 12 pages.

Deligne [6], *The Hodge Conjecture*, 5 pages.

Bombieri [2], *Problems of the Millennium: the Riemann Hypothesis*, 11 pages. Older tooling than the other four.

Jaffe and Witten [11], *Quantum Yang–Mills Theory*, 14 pages.

Wiles [21], *The Birch and Swinnerton-Dyer Conjecture*, 5 pages.

One page of Wiles is the largest shortfall here, because Birch and Swinnerton-Dyer is the problem the toolkit chapter recommends. That recommendation stands on an exact numerical identity, and the identity is not taken from this document. Whoever acts on the recommendation reads this first.

Wilkins [22], Riemann’s memoir translated by David R. Wilkins, *On the Number of Prime Numbers less than a Given Quantity*, 10 pages. The year is the translation’s, printed there as a preliminary version of December 1998; the memoir’s date, November 1859, is printed beside it. English, legible, and the practical way into the 1859 content.

5.4 NO TEXT TO READ

Riemann 1859 [19], handwritten working manuscript, *Nachlass* Cod. Ms. Riemann 3, *Rechen-seiten*. Four pages with zero extractable characters, counted page by page, rendered and read as images, the only way they can be read. The year on the entry is the manuscript’s; the scan dates from 2015. Its filename names the published memoir and it holds a draft, and the corpus chapter states what a citation built from that filename would do.

5.5 NOT A DOCUMENT

Clay Mathematics Institute [3]. Two facts in this research paper describe what that website says on 11 September 2026. The Millennium Problems index lists five unsolved, one solved, and Navier–Stokes in neither list. The Navier–Stokes page itself carries the label *Active*. The two do not agree. Both are recorded as statements about how a website sorts its own contents on one day at one address, and neither is used as a statement about a problem’s status.

5.6 CITED FOR ONE FACT EACH

The computations of Riemann zeros [10, 12, 16], the Birch and Swinnerton-Dyer computation [1], the curve tables [5], Dokchitser’s algorithm [7] and the press report on the prize [20] are each cited for the single fact the toolkit or Navier–Stokes chapter takes from them, as confirmed against the publisher or an index, and not for their content beyond it.

5.7 NAMED IN THESE SOURCES AND NOT READ HERE

Reference lists carry names, and this research paper repeats a handful at second hand. Leray on global weak solutions. Caffarelli, Kohn and Nirenberg on partial regularity. The Beale–Kato–Majda continuation criterion. Buckmaster and Vicol on non-uniqueness, then Albritton, Brué and Colombo on the same. Elgindi on axisymmetric blowup.

None is read, and none is cited here as established. Each has the status of a thing a read source mentions, which is all it has earned. An entry for any of them would assert a reading that is not done.

5.8 AVAILABILITY

The Clay statements, the two papers and the Lean repository are public at the addresses in the bibliography. The arbitrary-precision arithmetic behind the toolkit chapter is described in the instruments research paper [18].

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